

wheelers. Now, technology is an enabler and not a solution by itself. There is no scientific investigation of accidents in our country. Perhaps not even a single accident is being investigated scientifically because we blame everything on the driver. So, vehicle technologies, warning systems, and cameras are all very good and we need them.

Durgadutt Nedungadi:

I am in complete agreement that technology is an enabler and needs to be used to empower the drivers to driver better, give them tips on what aspects of their driving needs improvement as well as reward them for good driving behaviour. So just forcing down technology on to them, I believe, is counter effective.

We did a study with Zoom cars, where we installed 500 devices on their vehicles, and this study was conducted by IISC Bangalore. We found that without actually training the drivers on what our device was doing, without enabling the driver monitoring features for privacy reasons, just by having in-cab alerts stating that your driving skills are going wrong in some way, we actually saw that accidents reduced by almost 15%. So, enabling the drivers to behave better on the roads, to take the required precautions while driving goes a long way. And, vision based AI, ML technologies are doing exactly that today.

What can the Indian government, road safety authority, transport associations, and commercial fleet operators do to accelerate the pace of the adoption of this technology?

Ramesh Agarwal: We must first find out why such accidents take place and stop blaming only the drivers. As per reports, 1,38,000 people die because of accidents, out of which 26,400 die because the driver has fallen asleep. Now, we need to understand why drivers feel sleepy while driving. When truck drivers driving on national highways feel sleepy and seek rest, there are hardly any amenities to address driver fatigue. We addressed this by building a free of cost rest stop between Jaipur and Ajmer, called Nidra Daan. This rest stop accommodates 500 beds with blankets, has bathrooms with washing machines, and much more.

When we are talking about commercial vehicles and road transport operators, where does the Indian industry stand in curbing road accidents? How can the different industry segments contribute to the Union Government's mission of ensuring 50% reduction in road accident deaths by 2030?

Rohit Baluja: Today, the road accident investigation is led by the belief that it is the heavier vehicle driver's fault. You can see that the truck drivers are cautious drivers. Technology can help with driver

training, and we need more kinds of technology in the driver's cabin – where there is no air conditioning and isn't ergonomic. Technology should tell us what is going wrong and what are the needs of the drivers, etc. Also, we need to have data for deaths caused by road accidents. For example, in Delhi, 50% of deaths are of pedestrians. So, we need to know what the problem is and how best to address it using our actual intelligence and artificial intelligence.

How does technology help in training drivers?

Durgadutt Nedungadi: Through an advanced notification, technology can alert the driver that he is going to fall asleep; therefore, he needs to be careful. The fact that we can alert a driver when his eyes are droopy and get him to wake up and act, can save lives. We must use technology along with workforce development. Even 2 or 3% reduction in accident-causing incidents can lead to a great number of lives being saved. Camera-based technology like ours can give the driver, on his mobile phone app, a video of what exactly transpired within 2 minutes of the incident's occurrence. This gives him evidence for exoneration when he is not at fault. So, from all views, it can only make things better, rather than being an impediment.

Specifically in India, I feel, while there is a lot of training and change required in the process of training drivers prior to them getting their license, what technology can do today is to help coach them while they are on their day-to-day job to improve their driving behaviour. For example – if a fleet has 100 trucks with 150 drivers driving those trucks, on a week-to-week basis the fleet managers can track which drivers are making what kind of errors and develop a trend on who are the top 10 drivers who are driving well and thus need to be rewarded and who are the ones who need coaching to improve. Technology today enables us to use the driving videos of the driver to coach them better. Imbibing this culture of improving commercial vehicle driver behaviour in India will go a long way in ensuring that road safety norms are followed voluntarily as a matter of self-discipline than forceful law enforcement in the long run.

The national growth safety policy of 2010 stipulates that the government will establish a road safety information database to help states, union territories, and local bodies, improve the quality of crash information and data collection and transmission. How is the industry really supporting the government?

Durgadutt Nedungadi: The investigations that happen on crash sites are not very scientific. If you look at places like the UK, the police have forensic investigators who are specially trained to inspect

vehicles after an accident. Now it will take us a long time to get there. But at least if we have footage of what happened on the road, and what could have been done to prevent that event, then at least it's a starting point. And as tech companies, we are willing to step up and provide whatever is required from our end. So, while I think that the vision is the right one, there are several steps that need to be taken to make it an executable plan on the ground.

In what ways can the government support the road safety initiative?

Durgadutt Nedungadi: Firstly, I think there needs to be a concerted effort to make our roads infrastructure consistent. What I mean by that is, that it should not take us long to develop standard road signs and traffic lights for having predictability in the way we define lanes on our roads. Once we have that consistency, it allows technology to use that consistency to improve safety. That is step 1. Step 2 of course is, I would say an encouragement to fleets to increasingly use technology to save lives, legislation, etc.

Are there any cases, whether in India or abroad, of something that's worked well or not?

Durgadutt Nedungadi: There are several stories where technology has worked dramatically. One organisation found that, with just the implementation of this device on 400 of their vehicles, the alert notifications came down from 1.2 alerts for every 1.5 hours of driving to 0.08 alerts at the same time. This reflects the improvement that they've seen, and it directly correlates to driver behaviour, safety, and accident reduction.

Is there any city in India that can be considered the gold standard in road safety?

Rohit Baluja: Chandigarh is a perfect example because of the environment, which is quite conducive. Technology today is required; we cannot do without technology for road maintenance, driver training, and data collection. As an enabler, technology is good, but we need to match it with actual intelligence.

As statistics suggest, the way forward in increasing road safety is to implement AI-based and vision-based technologies in fleet vehicles. As Durgadutt rightly says, "There is ancillary work that needs to be done so that technology can work. I believe that the faster we adapt, the quicker the difference we make towards getting to our goal. In India, we are seeing an uptake of interest across segments, which is really encouraging because that also means that drivers themselves see the benefit in being part of this whole exercise of using that technology to improve drive time."